

CommunityHub

Home → Login → Dashboard → Forums / Events / Members / Groups

CS5450 Mobile Programming · Lakehead University

A React Native / Expo community portal backed by Firebase Authentication and Cloud Firestore. Users register with email and password, get a personalized dashboard, post in discussion forums, join events, browse members, and join interest groups. Features real-time data sync, role-based access (admin / member), category filters, and a cross-platform responsive dark UI.

■ 5 Screens ■ Firebase Auth ■ Firestore DB ■ Real-time Sync ■ Role-based Access ■ Android & Web

1. Features

- User Registration & Login with Firebase Authentication (email/password), admin and member roles
- Personalized Dashboard showing your post count, joined events, groups, and recent activity
- Discussion Forums — create posts, reply in threads, like posts with real-time Firestore sync
- Event Calendar — create events, join/leave with RSVP count, filter by category
- Member Directory — search members by name or interest, view full profile modal
- Interest Groups — join/leave groups, live member count, activity feed per group
- Global search bar on every screen — searches forums, events, members, and groups
- Firestore Security Rules enforce role-based read/write access
- Works on Android emulator (Expo Go), physical device, and Chrome/Web

2. Project Structure

```
community-hub/
|
+-- App.js Entry point. NavigationContainer + Bottom Tab + Stack navigators
+-- index.js Expo registerRootComponent entry for web
+-- app.json Expo config: name, icons, splash, bundle IDs, web bundler
+-- package.json npm dependencies and run scripts
+-- firestore.rules Firestore security rules (role-based access control)
+-- README.pdf This document
|
+-- src/
|   +-- hooks/
|       | +-- useAuth.js AuthContext + useAuth hook (Firebase auth state listener)
|       |
|       +-- services/
|           | +-- firebase.js Firebase app init, exports auth and db
|           | +-- authService.js register, login, logout, fetchUserProfile
```

```
| | +-- firestoreService.js All Firestore CRUD + real-time onSnapshot listeners
| |
| +-- screens/
| | +-- AuthScreen.js Login + Register UI with Firebase Auth
| | +-- DashboardScreen.js Personalized home with stats, events, groups
| | +-- ForumsScreen.js Forum list + ThreadDetail with replies and likes
| | +-- EventsScreen.js Events list + create event + join/leave
| | +-- MembersScreen.js Member directory + Groups screen
| |
| +-- utils/
| +-- helpers.js Category colours, avatarBg, formatDate, timeAgo
```

3. Requirements

Software Needed

Tool	Version / Detail
Android Studio	Hedgehog 2023.1.1 or newer
Node.js	v18 LTS or newer — https://nodejs.org
npm	Included with Node.js (v10+)
Expo CLI	Via npx — no global install required
Expo SDK	51 (react-native 0.74)
Firebase	v10.12.2 (Auth + Firestore)
Compile SDK	34 (Android 14)
Min SDK	21 (Android 5.0+)

4. How to Run

Step 1 — Open the Project

Extract the ZIP file. Open a Command Prompt and navigate to the project folder:

```
cd C:\Users\vahor\AndroidStudioProjects\Group_9_Challenge_3\community-hub
```

Step 2 — Install Dependencies

Run npm install to download all packages listed in package.json:

```
npm install
```

Step 3 — Start the Emulator

Open Android Studio → Tools → Device Manager → click  Play next to your AVD (e.g. Pixel 6 API 34). Wait until the Android home screen appears.

Step 4 — Set ADB Path

Run this in CMD so Expo can find the Android emulator:

```
set PATH=%PATH%;C:\Users\vahor\AppData\Local\Android\Sdk\platform-tools
```

Step 5 — Start Expo Dev Server

```
npx expo start
```

Step 6 — Open on Android

Press a in the terminal. Expo installs Expo Go on the emulator and launches the app. If the URL prompt appears in Expo Go, type:

```
exp://127.0.0.1:8081
```

Run on Chrome (no emulator needed)

Press w in the terminal after starting, or run:

```
npx expo start --web
```

5. Firebase Configuration

Setting	Value
Project name	community-hub
Project ID	community-hub-7a150
Auth domain	community-hub-7a150.firebaseio.com
Region	northamerica-northeast1 (Montreal)
Auth method	Email / Password
Database	Cloud Firestore — Standard edition
Config file	src/services/firebase.js

6. File Descriptions

File	Purpose
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App.js	Entry point. Sets up NavigationContainer, Bottom Tab Navigator with 5 tabs, and auth gate
index.js	Registers root component for Expo web bundler using registerRootComponent
firebase.js	Initialises Firebase app with project config. Exports auth and db instances
authService.js	Wraps Firebase Auth — registerUser, loginUser, logoutUser, fetchUserProfile
firestoreService.js	All Firestore CRUD operations and real-time onSnapshot listeners for forums, events, groups
useAuth.js	React Context + hook providing user and profile state to all screens
AuthScreen.js	Login and Register forms with Firebase Auth, error handling, and role assignment
DashboardScreen.js	Personalized home showing stats, recent forum posts, upcoming events, and joined groups
ForumsScreen.js	Forum post list with category filter, create post modal, and ThreadDetail with replies
EventsScreen.js	Event cards with join/leave, create event modal, date badge, and category filter
MembersScreen.js	Member directory grid with search and profile modal. Also contains GroupsScreen
helpers.js	Utility functions: CAT_COLORS map, avatarBg colour picker, formatDate, timeAgo
firestore.rules	Production Firestore security rules — role-based read/write/delete permissions
app.json	Expo config: app name, slug, version, icon, splash screen, Android package name
package.json	Lists all npm dependencies including expo, firebase, react-navigation, and react-native

7. Technologies Used

Technology	Purpose
React Native / Expo	Cross-platform mobile UI framework. Expo SDK 51 for managed workflow on Android, iOS, and Web
Firebase Auth	Email/password authentication with role assignment stored in Firestore users collection
Cloud Firestore	NoSQL real-time database. onSnapshot listeners update UI instantly on data changes
React Navigation 6	Bottom tab navigator for main screens + native stack for forum thread detail
JavaScript ES2022	Primary language. Async/await for all Firebase calls, destructuring, and arrow functions
React Context API	AuthProvider wraps the app, exposes user and profile to all screens via useAuth hook
StyleSheet API	All styles defined with React Native StyleSheet.create for optimised rendering
FlatList / ScrollView	Efficient lazy rendering of forum posts, event cards, member grid, and group list
SafeAreaView	Ensures UI avoids device notches and status bars on Android and iOS

8. Future Enhancements

- Add profile photo upload using Firebase Storage
- Push notifications for new forum replies and event reminders
- Direct messaging between community members
- Dark/light theme toggle in app settings
- Pagination for forum posts and member directory using Firestore cursors

- Event map view showing locations using react-native-maps
- Admin dashboard for managing users, posts, and events
- Export community data to CSV for reporting

9. Submission Checklist (D2L ZIP)

Item	Description	Status
src/ folder	All JS source files (screens, services, hooks, utils)	✓ Required
App.js	Root navigation entry point	✓ Required
index.js	Expo web entry point	✓ Required
app.json	Expo app configuration	✓ Required
package.json	npm dependency manifest	✓ Required
firestore.rules	Firestore security rules	✓ Required
README.pdf	This document	✓ Required
Firebase config	Real config in src/services/firebase.js	✓ Required
GitHub URL	https://github.com/mvahora217/community-hub-cs5450	✓ Required